Messaging and stream processing

Faculty of Informatics, Juraj Dobrila University of Pula

assoc. prof. Nikola Tankovic, PhD

Sample chapter for the proposed course plan.

1 Learning outcome

Explain why a message consumer may see the same event twice.

Prerequisites: Lectures 03, 05, and 08.

2 Core idea

A message broker decouples the time at which a producer sends work from the time at which a consumer processes it. Under at-least-once delivery, a consumer must be prepared to receive a message more than once.

3 Worked scenario

A notification consumer sends an email but loses its acknowledgement. The broker redelivers the confirmation event.

4 Discussion

Where could duplicate detection live, and what would it need to remember?

5 Recap

A delivery guarantee must be connected to the effects performed by the consumer.